Manage FPolicy event configuration

ONTAP 9.14.1 REST API reference

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Table of Contents

Manage FPolicy event configuration . . . 1
Protocols fpolicy svm.uuid events endpoint overview . . . 1
Retrieve the FPolicy event configuration for all events for an SVM . . . 10
Create the FPolicy event configuration for an SVM . . . 22
Delete a specific FPolicy event configuration for an SVM . . . 31
Retrieve a specific FPolicy event configuration for an SVM . . . 33
Update a specific FPolicy event configuration for an SVM . . . 39

Manage FPolicy event configuration

Protocols fpolicy svm.uuid events endpoint overview

Overview

FPolicy events configurations allow you to specify which file access is monitored. As part of an FPolicy event, you can configure the SVM for which the events are generated, the name of the event configuration, the protocol (cifs, nfsv3/nfsv4) for which the events are generated, the file operations which are monitored, and filters that can be used to filter the unwanted notification generation for a specified protocol and file operation.

Each protocol has a set of supported file operations and filters. An SVM can have multiple events. A single FPolicy policy can have multiple FPolicy events.

FPolicy events can also be configured to monitor file operations which fail due to lack of permissions. You can specify which file operation to monitor for failure. However, filters can not be used to filter failed file operations.

Examples

Creating an FPolicy event for a CIFS protocol with all the supported file operations and filters

```
# The API:
POST /protocols/fpolicy/{svm.uuid}/events

# The call:
curl -X POST "https://<mgmt-ip>/api/protocols/fpolicy/4f643fb4-fd21-11e8-
ae49-0050568e2c1e/events?return_records=true" -H "accept:
application/json" -H "Content-Type: application/json" -d "{
\"file_operations\": { \"close\": true, \"create\": true, \"create_dir\":
true, \"delete\": true, \"delete_dir\": true, \"getattr\": true, \"open\":
true, \"read\": true, \"rename\": true, \"rename_dir\": true, \"setattr\":
true, \"write\": true }, \"filters\": { \"close_with_modification\": true,
\"close_with_read\": true, \"close_without_modification\": true,
\"first_read\": true, \"first_write\": true, \"monitor_ads\": true,
\"offline_bit\": true, \"open_with_delete_intent\": true,
\"open_with_write_intent\": true, \"write_with_size_change\": true },
\"name\": \"event_cifs\", \"protocol\": \"cifs\", \"volume_monitoring\":
true}"

# The response:
{
"num_records": 1,
  "records": [
    {
      "name": "event_cifs",
```

```
      "protocol": "cifs",
      "volume_monitoring": true,
      "monitor_fileop_failure": false,
      "file_operations": {
        "close": true,
        "create": true,
        "create_dir": true,
        "delete": true,
        "delete_dir": true,
        "getattr": true,
        "open": true,
        "read": true,
        "write": true,
        "rename": true,
        "rename_dir": true,
        "setattr": true
      },
      "filters": {
        "monitor_ads": true,
        "close_with_modification": true,
        "close_without_modification": true,
        "close_with_read": true,
        "first_read": true,
        "first_write": true,
        "offline_bit": true,
        "open_with_delete_intent": true,
        "open_with_write_intent": true,
        "write_with_size_change": true
      }
    }
  ]
}
```

Creating an FPolicy event for an NFS protocol with all the supported file operations and filters

---

```
# The API:
post /protocols/fpolicy/{svm.uuid}/events

# The call:
curl -X POST "https://<mgmt-ip>/api/protocols/fpolicy/4f643fb4-fd21-11e8-
ae49-0050568e2c1e/events?return_records=true" -H "accept:
application/json" -H "Content-Type: application/json" -d "{
\"file_operations\": { \"create\": true, \"create_dir\": true, \"delete\":
true, \"delete_dir\": true, \"link\": true, \"lookup\": true, \"read\":
true, \"rename\": true, \"rename_dir\": true, \"setattr\": true,
\"symlink\": true, \"write\": true }, \"filters\": { \"offline_bit\":
true, \"write_with_size_change\": true }, \"name\": \"event_nfsv3\",
\"protocol\": \"nfsv3\", \"volume_monitoring\": false}"

# The response:
{
"num_records": 1,
"records": [
  {
    "name": "event_nfsv3",
    "protocol": "nfsv3",
    "volume_monitoring": false,
    "monitor_fileop_failure": false,
    "file_operations": {
      "create": true,
      "create_dir": true,
      "delete": true,
      "delete_dir": true,
      "link": true,
      "lookup": true,
      "read": true,
      "write": true,
      "rename": true,
      "rename_dir": true,
      "setattr": true,
      "symlink": true
    },
    "filters": {
    "offline_bit": true,
    "write_with_size_change": true
    }
  }
]
}
```

Creating an FPolicy event to monitor failed file operations for an NFS protocol with all the supported file operations

```
# The API:
post /protocols/fpolicy/{svm.uuid}/events

# The call:
curl -X POST "https://<mgmt-ip>/api/protocols/fpolicy/b5087518-40b3-11ed-
b3eb-005056bbe901/events?return_records=false" -H "accept:
application/json" -H  "authorization: Basic YWRtaW46bmV0YXBwMSE=" -H
"Content-Type: application/json" -d "{  \"file_operations\": {
\"create\": true,    \"create_dir\": true,    \"delete\": true,
\"delete_dir\": true,    \"link\": true,    \"read\": true,    \"rename\":
true,    \"rename_dir\": true,    \"write\": true  },  \"name\":
\"nfs_failed_op\",  \"protocol\": \"nfsv3\",  \"monitor_fileop_failure\":
true,  \"volume_monitoring\": false}"

# The response:
{
"num_records": 1,
"records": [
  {
    "name": "nfs_failed_op",
    "protocol": "nfsv3",
    "volume_monitoring": false,
    "monitor_fileop_failure": true,
    "file_operations": {
      "create": true,
      "create_dir": true,
      "delete": true,
      "delete_dir": true,
      "link": true,
      "read": true,
      "write": true,
      "rename": true,
      "rename_dir": true
    }
  }
]
}
```

Retrieving all of the FPolicy event configurations configured to monitor failed file operations for a specified SVM

```
# The API:
post /protocols/fpolicy/{svm.uuid}/events

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/fpolicy/b5087518-40b3-11ed-
b3eb-
005056bbe901/events?monitor_fileop_failure=true&fields=*&return_records=tr
ue&return_timeout=15" -H  "accept: application/json"

# The response:
{
"records": [
  {
    "svm": {
    "uuid": "b5087518-40b3-11ed-b3eb-005056bbe901"
    },
    "name": "fo_event",
    "protocol": "cifs",
    "volume_monitoring": false,
    "monitor_fileop_failure": true,
    "file_operations": {
      "close": false,
      "create": false,
      "create_dir": false,
      "delete": false,
      "delete_dir": false,
      "getattr": false,
      "link": false,
      "lookup": false,
      "open": true,
      "read": false,
      "write": false,
      "rename": false,
      "rename_dir": false,
      "setattr": false,
      "symlink": false
    }
  }
],
"num_records": 1
}
```

Retrieving all of the FPolicy event configurations for a specified SVM

```
# The API:
GET /protocols/fpolicy/{svm.uuid}/events

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/fpolicy/4f643fb4-fd21-11e8-
ae49-0050568e2c1e/events/?fields=*&return_records=true&return_timeout=15"
-H "accept: application/json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "4f643fb4-fd21-11e8-ae49-0050568e2c1e"
    },
    "name": "cluster",
    "protocol": "cifs",
    "volume_monitoring": false,
    "monitor_fileop_failure": false,
    "file_operations": {
      "close": true,
      "create": false,
      "create_dir": false,
      "delete": false,
      "delete_dir": false,
      "getattr": false,
      "link": false,
      "lookup": false,
      "open": false,
      "read": false,
      "write": false,
      "rename": false,
      "rename_dir": false,
      "setattr": false,
      "symlink": false
    },
    "filters": {
      "monitor_ads": false,
      "close_with_modification": false,
      "close_without_modification": false,
      "close_with_read": true,
      "first_read": false,
```

```
      "first_write": false,
      "offline_bit": false,
      "open_with_delete_intent": false,
      "open_with_write_intent": false,
      "write_with_size_change": false,
      "setattr_with_owner_change": false,
      "setattr_with_group_change": false,
      "setattr_with_sacl_change": false,
      "setattr_with_dacl_change": false,
      "setattr_with_modify_time_change": false,
      "setattr_with_access_time_change": false,
      "setattr_with_creation_time_change": false,
      "setattr_with_mode_change": false,
      "setattr_with_size_change": false,
      "setattr_with_allocation_size_change": false,
      "exclude_directory": false
    }
  },
  {
    "svm": {
      "uuid": "4f643fb4-fd21-11e8-ae49-0050568e2c1e"
    },
    "name": "event_cifs",
    "protocol": "cifs",
    "volume_monitoring": true,
    "monitor_fileop_failure": false,
    "file_operations": {
      "close": true,
      "create": true,
      "create_dir": true,
      "delete": true,
      "delete_dir": true,
      "getattr": true,
      "link": false,
      "lookup": false,
      "open": true,
      "read": true,
      "write": true,
      "rename": true,
      "rename_dir": true,
      "setattr": true,
      "symlink": false
    },
    "filters": {
      "monitor_ads": true,
      "close_with_modification": true,
```

```
      "close_without_modification": true,
      "close_with_read": true,
      "first_read": true,
      "first_write": true,
      "offline_bit": true,
      "open_with_delete_intent": true,
      "open_with_write_intent": true,
      "write_with_size_change": true,
      "setattr_with_owner_change": false,
      "setattr_with_group_change": false,
      "setattr_with_sacl_change": false,
      "setattr_with_dacl_change": false,
      "setattr_with_modify_time_change": false,
      "setattr_with_access_time_change": false,
      "setattr_with_creation_time_change": false,
      "setattr_with_mode_change": false,
      "setattr_with_size_change": false,
      "setattr_with_allocation_size_change": false,
      "exclude_directory": false
    }
  }
],
"num_records": 2
}
```

Retrieving a specific FPolicy event configuration for an SVM

```
# The API:
GET /protocols/fpolicy/{svm.uuid}/events/{name}

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/fpolicy/4f643fb4-fd21-11e8-
ae49-
0050568e2c1e/events/event_cifs?fields=*&return_records=true&return_timeout
=15" -H "accept: application/json"

# The response:
{
"svm": {
  "uuid": "4f643fb4-fd21-11e8-ae49-0050568e2c1e"
},
"name": "event_cifs",
```

```
"protocol": "cifs",
"volume_monitoring": true,
"monitor_fileop_failure": false,
"file_operations": {
  "close": true,
  "create": true,
  "create_dir": true,
  "delete": true,
  "delete_dir": true,
  "getattr": true,
  "link": false,
  "lookup": false,
  "open": true,
  "read": true,
  "write": true,
  "rename": true,
  "rename_dir": true,
  "setattr": true,
  "symlink": false
},
"filters": {
  "monitor_ads": true,
  "close_with_modification": true,
  "close_without_modification": true,
  "close_with_read": true,
  "first_read": true,
  "first_write": true,
  "offline_bit": true,
  "open_with_delete_intent": true,
  "open_with_write_intent": true,
  "write_with_size_change": true,
  "setattr_with_owner_change": false,
  "setattr_with_group_change": false,
  "setattr_with_sacl_change": false,
  "setattr_with_dacl_change": false,
  "setattr_with_modify_time_change": false,
  "setattr_with_access_time_change": false,
  "setattr_with_creation_time_change": false,
  "setattr_with_mode_change": false,
  "setattr_with_size_change": false,
  "setattr_with_allocation_size_change": false,
  "exclude_directory": false
}
}
```

Updating a specific FPolicy event configuration for a specified SVM

```
# The API:
PATCH /protocols/fpolicy/{svm.uuid}/events/{name}

# The call:
curl -X PATCH "https://<mgmt-ip>/api/protocols/fpolicy/4f643fb4-fd21-11e8-
ae49-0050568e2c1e/events/event_cifs" -H "accept: application/json" -H
"Content-Type: application/json" -d "{ \"file_operations\": { \"close\":
false, \"create\": false, \"read\": true }, \"filters\": {
\"close_with_modification\": false, \"close_with_read\": false,
\"close_without_modification\": false }, \"protocol\": \"cifs\",
\"volume_monitoring\": false}"
```

Deleting a specific FPolicy event configuration for a specific SVM

```
# The API:
DELETE /protocols/fpolicy/{svm.uuid}/events/{name}

# The call:
curl -X DELETE "https://<mgmt-ip>/api/protocols/fpolicy/4f643fb4-fd21-
11e8-ae49-0050568e2c1e/events/event_cifs" -H "accept: application/json"
```

Retrieve the FPolicy event configuration for all events for an SVM

GET `/protocols/fpolicy/{svm.uuid}/events`

Introduced In: 9.6

Retrieves FPolicy event configurations for all events for a specified SVM. ONTAP allows the creation of cluster-level FPolicy events that act as a template for all the data SVMs belonging to the cluster. These cluster-level FPolicy events are also retrieved for the specified SVM.

Related ONTAP commands

- `fpolicy policy event show`

Learn more

- DOC /protocols/fpolicy/{svm.uuid}/events

Parameters

Name	Type	In	Required	Description
protocol	string	query	False	Filter by protocol
monitor_fileop_failure	boolean	query	False	Filter by monitor_fileop_failure • Introduced in: 9.13
name	string	query	False	Filter by name
filters.exclude_directory	boolean	query	False	Filter by filters.exclude_directory
filters.write_with_size_change	boolean	query	False	Filter by filters.write_with_size_change
filters.monitor_ads	boolean	query	False	Filter by filters.monitor_ads
filters.setattr_with_dacl_change	boolean	query	False	Filter by filters.setattr_with_dacl_change
filters.offline_bit	boolean	query	False	Filter by filters.offline_bit
filters.setattr_with_creation_time_change	boolean	query	False	Filter by filters.setattr_with_creation_time_change
filters.open_with_delete_intent	boolean	query	False	Filter by filters.open_with_delete_intent
filters.first_read	boolean	query	False	Filter by filters.first_read

Name	Type	In	Required	Description
filters.setattr_with_mode_change	boolean	query	False	Filter by filters.setattr_with_mode_change
filters.close_with_modification	boolean	query	False	Filter by filters.close_with_modification
filters.first_write	boolean	query	False	Filter by filters.first_write
filters.close_without_modification	boolean	query	False	Filter by filters.close_without_modification
filters.setattr_with_access_time_change	boolean	query	False	Filter by filters.setattr_with_access_time_change
filters.setattr_with_modify_time_change	boolean	query	False	Filter by filters.setattr_with_modify_time_change
filters.setattr_with_owner_change	boolean	query	False	Filter by filters.setattr_with_owner_change
filters.setattr_with_size_change	boolean	query	False	Filter by filters.setattr_with_size_change
filters.setattr_with_allocation_size_change	boolean	query	False	Filter by filters.setattr_with_allocation_size_change
filters.close_with_read	boolean	query	False	Filter by filters.close_with_read
filters.setattr_with_group_change	boolean	query	False	Filter by filters.setattr_with_group_change
filters.open_with_write_intent	boolean	query	False	Filter by filters.open_with_write_intent

Name	Type	In	Required	Description
filters.setattr_with_sacl_change	boolean	query	False	Filter by filters.setattr_with_sacl_change
file_operations.read	boolean	query	False	Filter by file_operations.read
file_operations.rename_dir	boolean	query	False	Filter by file_operations.rename_dir
file_operations.access	boolean	query	False	Filter by file_operations.access • Introduced in: 9.13
file_operations.delete_dir	boolean	query	False	Filter by file_operations.delete_dir
file_operations.create_dir	boolean	query	False	Filter by file_operations.create_dir
file_operations.open	boolean	query	False	Filter by file_operations.open
file_operations.create	boolean	query	False	Filter by file_operations.create
file_operations.write	boolean	query	False	Filter by file_operations.write
file_operations.close	boolean	query	False	Filter by file_operations.close
file_operations.getattr	boolean	query	False	Filter by file_operations.getattr
file_operations.setattr	boolean	query	False	Filter by file_operations.setattr

Name	Type	In	Required	Description
file_operations.delete	boolean	query	False	Filter by file_operations.delete
file_operations.rename	boolean	query	False	Filter by file_operations.rename
file_operations.link	boolean	query	False	Filter by file_operations.link
file_operations.lookup	boolean	query	False	Filter by file_operations.lookup
file_operations.symlink	boolean	query	False	Filter by file_operations.symlink
volume_monitoring	boolean	query	False	Filter by volume_monitoring
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[fpolicy_event]	

Example response

```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "name": "event_cifs",
    "protocol": "cifs",
    "svm": {
      "uuid": "string"
    }
  }
}
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

file_operations

Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.

Name	Type	Description
access	boolean	Access operations
close	boolean	File close operations
create	boolean	File create operations
create_dir	boolean	Directory create operations
delete	boolean	File delete operations
delete_dir	boolean	Directory delete operations
getattr	boolean	Get attribute operations
link	boolean	Link operations
lookup	boolean	Lookup operations
open	boolean	File open operations
read	boolean	File read operations
rename	boolean	File rename operations
rename_dir	boolean	Directory rename operations

Name	Type	Description
setattr	boolean	Set attribute operations
symlink	boolean	Symbolic link operations
write	boolean	File write operations

filters

Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.

Name	Type	Description
close_with_modification	boolean	Filter the client request for close with modification.
close_with_read	boolean	Filter the client request for close with read.
close_without_modification	boolean	Filter the client request for close without modification.
exclude_directory	boolean	Filter the client requests for directory operations. When this filter is specified directory operations are not monitored.
first_read	boolean	Filter the client requests for the first-read.
first_write	boolean	Filter the client requests for the first-write.
monitor_ads	boolean	Filter the client request for alternate data stream.
offline_bit	boolean	Filter the client request for offline bit set. FPolicy server receives notification only when offline files are accessed.
open_with_delete_intent	boolean	Filter the client request for open with delete intent.
open_with_write_intent	boolean	Filter the client request for open with write intent.

Name	Type	Description
setattr_with_access_time_change	boolean	Filter the client setattr requests for changing the access time of a file or directory.
setattr_with_allocation_size_change	boolean	Filter the client setattr requests for changing the allocation size of a file.
setattr_with_creation_time_change	boolean	Filter the client setattr requests for changing the creation time of a file or directory.
setattr_with_dacl_change	boolean	Filter the client setattr requests for changing dacl on a file or directory.
setattr_with_group_change	boolean	Filter the client setattr requests for changing group of a file or directory.
setattr_with_mode_change	boolean	Filter the client setattr requests for changing the mode bits on a file or directory.
setattr_with_modify_time_change	boolean	Filter the client setattr requests for changing the modification time of a file or directory.
setattr_with_owner_change	boolean	Filter the client setattr requests for changing owner of a file or directory.
setattr_with_sacl_change	boolean	Filter the client setattr requests for changing sacl on a file or directory.
setattr_with_size_change	boolean	Filter the client setattr requests for changing the size of a file.
write_with_size_change	boolean	Filter the client request for write with size change.

svm

Name	Type	Description
uuid	string	SVM UUID

fpolicy_event

The information that a FPolicy process needs to determine what file access operations to monitor and for which of the monitored events notifications should be sent to the external FPolicy server.

Name	Type	Description
file_operations	file_operations	Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.
filters	filters	Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.
monitor_fileop_failure	boolean	Specifies whether failed file operations monitoring is required.
name	string	Specifies the name of the FPolicy event.
protocol	string	Protocol for which event is created. If you specify protocol, then you must also specify a valid value for the file operation parameters. The value of this parameter must be one of the following: • cifs - for the CIFS protocol. • nfsv3 - for the NFSv3 protocol. • nfsv4 - for the NFSv4 protocol.
svm	svm	
volume_monitoring	boolean	Specifies whether volume operation monitoring is required.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create the FPolicy event configuration for an SVM

POST `/protocols/fpolicy/{svm.uuid}/events`

Introduced In: 9.6

Creates an FPolicy event configuration for a specified SVM. FPolicy event creation is allowed only on data SVMs. When a protocol is specified, you must specify a file operation or a file operation and filters. When FPolicy event is configured to monitor failed file operations, you must specify protocol and file operations. Filters are not supported when failed file operations are monitored.

Required properties

- `svm.uuid` - Existing SVM in which to create the FPolicy event.
- `name` - Name of the FPolicy event.

Recommended optional properties

- `file-operations` - List of file operations to monitor.
- `protocol` - Protocol for which the file operations should be monitored.
- `filters` - List of filters for the specified file operations.
- `monitor-fileop-failure` - Enabled monitoring of failed file operations.

Default property values

If not specified in POST, the following default property values are assigned:

- `file_operations.*` - *false*
- `filters.*` - *false*
- `volume-monitoring` - *false*
- `monitor-fileop-failure` - *false*

Related ONTAP commands

- `fpolicy policy event create`

Learn more

- DOC /protocols/fpolicy/{svm.uuid}/events

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Request Body

Name	Type	Description
file_operations	file_operations	Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.
filters	filters	Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.
monitor_fileop_failure	boolean	Specifies whether failed file operations monitoring is required.

Name	Type	Description
name	string	Specifies the name of the FPolicy event.
protocol	string	Protocol for which event is created. If you specify protocol, then you must also specify a valid value for the file operation parameters. The value of this parameter must be one of the following: • cifs - for the CIFS protocol. • nfsv3 - for the NFSv3 protocol. • nfsv4 - for the NFSv4 protocol.
svm	svm	
volume_monitoring	boolean	Specifies whether volume operation monitoring is required.

Example request

```
{
  "name": "event_cifs",
  "protocol": "cifs",
  "svm": {
    "uuid": "string"
  }
}
```

Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[fpolicy_event]	

Example response

```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "name": "event_cifs",
    "protocol": "cifs",
    "svm": {
      "uuid": "string"
    }
  }
}
```

Headers

Name	Description	Type
Location	Useful for tracking the resource location	string

Error

```
Status: Default
```

Error Code	Description
9764929	The file operation is not supported by the protocol
9764955	The filter is not supported by the protocol
9764930	The filter is not supported by any of the file operations
9764946	The protocol is specified without a file operation or a file operation and filter pair

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

file_operations

Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.

Name	Type	Description
access	boolean	Access operations
close	boolean	File close operations
create	boolean	File create operations
create_dir	boolean	Directory create operations
delete	boolean	File delete operations
delete_dir	boolean	Directory delete operations
getattr	boolean	Get attribute operations
link	boolean	Link operations
lookup	boolean	Lookup operations
open	boolean	File open operations
read	boolean	File read operations
rename	boolean	File rename operations
rename_dir	boolean	Directory rename operations
setattr	boolean	Set attribute operations
symlink	boolean	Symbolic link operations
write	boolean	File write operations

filters

Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.

Name	Type	Description
close_with_modification	boolean	Filter the client request for close with modification.
close_with_read	boolean	Filter the client request for close with read.
close_without_modification	boolean	Filter the client request for close without modification.
exclude_directory	boolean	Filter the client requests for directory operations. When this filter is specified directory operations are not monitored.
first_read	boolean	Filter the client requests for the first-read.
first_write	boolean	Filter the client requests for the first-write.
monitor_ads	boolean	Filter the client request for alternate data stream.
offline_bit	boolean	Filter the client request for offline bit set. FPolicy server receives notification only when offline files are accessed.
open_with_delete_intent	boolean	Filter the client request for open with delete intent.
open_with_write_intent	boolean	Filter the client request for open with write intent.
setattr_with_access_time_change	boolean	Filter the client setattr requests for changing the access time of a file or directory.
setattr_with_allocation_size_change	boolean	Filter the client setattr requests for changing the allocation size of a file.
setattr_with_creation_time_change	boolean	Filter the client setattr requests for changing the creation time of a file or directory.

Name	Type	Description
setattr_with_dacl_change	boolean	Filter the client setattr requests for changing dacl on a file or directory.
setattr_with_group_change	boolean	Filter the client setattr requests for changing group of a file or directory.
setattr_with_mode_change	boolean	Filter the client setattr requests for changing the mode bits on a file or directory.
setattr_with_modify_time_change	boolean	Filter the client setattr requests for changing the modification time of a file or directory.
setattr_with_owner_change	boolean	Filter the client setattr requests for changing owner of a file or directory.
setattr_with_sacl_change	boolean	Filter the client setattr requests for changing sacl on a file or directory.
setattr_with_size_change	boolean	Filter the client setattr requests for changing the size of a file.
write_with_size_change	boolean	Filter the client request for write with size change.

svm

Name	Type	Description
uuid	string	SVM UUID

fpolicy_event

The information that a FPolicy process needs to determine what file access operations to monitor and for which of the monitored events notifications should be sent to the external FPolicy server.

Name	Type	Description
file_operations	file_operations	Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.
filters	filters	Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.
monitor_fileop_failure	boolean	Specifies whether failed file operations monitoring is required.
name	string	Specifies the name of the FPolicy event.
protocol	string	Protocol for which event is created. If you specify protocol, then you must also specify a valid value for the file operation parameters. The value of this parameter must be one of the following: • cifs - for the CIFS protocol. • nfsv3 - for the NFSv3 protocol. • nfsv4 - for the NFSv4 protocol.
svm	svm	
volume_monitoring	boolean	Specifies whether volume operation monitoring is required.

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Delete a specific FPolicy event configuration for an SVM

DELETE `/protocols/fpolicy/{svm.uuid}/events/{name}`

Introduced In: 9.6

Deletes a specific FPolicy event configuration for an SVM. A cluster-level FPolicy event configuration cannot be modified for a data SVM through REST. An FPolicy event that is attached to an FPolicy policy cannot be deleted.

Related ONTAP commands

- `fpolicy policy event delete`

Learn more

- DOC /protocols/fpolicy/{svm.uuid}/events

Parameters

Name	Type	In	Required	Description
name	string	path	True	
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

Error Code	Description
9764874	The FPolicy event is a cluster event
9764947	The FPolicy event is attached to an FPolicy policy

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve a specific FPolicy event configuration for an SVM

GET `/protocols/fpolicy/{svm.uuid}/events/{name}`

Introduced In: 9.6

Retrieves a specific FPolicy event configuration for an SVM. A cluster-level FPolicy event configuration cannot be retrieved for a data SVM through a REST API.

Related ONTAP commands

- `fpolicy policy event show`

Learn more

- DOC /protocols/fpolicy/{svm.uuid}/events

Parameters

Name	Type	In	Required	Description
name	string	path	True	

Name	Type	In	Required	Description
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
file_operations	file_operations	Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.
filters	filters	Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.
monitor_fileop_failure	boolean	Specifies whether failed file operations monitoring is required.
name	string	Specifies the name of the FPolicy event.
protocol	string	Protocol for which event is created. If you specify protocol, then you must also specify a valid value for the file operation parameters. The value of this parameter must be one of the following: • cifs - for the CIFS protocol. • nfsv3 - for the NFSv3 protocol. • nfsv4 - for the NFSv4 protocol.
svm	svm	

Name	Type	Description
volume_monitoring	boolean	Specifies whether volume operation monitoring is required.

Example response

```
{
  "name": "event_cifs",
  "protocol": "cifs",
  "svm": {
    "uuid": "string"
  }
}
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

file_operations

Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.

Name	Type	Description
access	boolean	Access operations
close	boolean	File close operations
create	boolean	File create operations
create_dir	boolean	Directory create operations
delete	boolean	File delete operations
delete_dir	boolean	Directory delete operations
getattr	boolean	Get attribute operations
link	boolean	Link operations
lookup	boolean	Lookup operations
open	boolean	File open operations
read	boolean	File read operations
rename	boolean	File rename operations
rename_dir	boolean	Directory rename operations
setattr	boolean	Set attribute operations
symlink	boolean	Symbolic link operations
write	boolean	File write operations

filters

Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.

Name	Type	Description
close_with_modification	boolean	Filter the client request for close with modification.
close_with_read	boolean	Filter the client request for close with read.
close_without_modification	boolean	Filter the client request for close without modification.
exclude_directory	boolean	Filter the client requests for directory operations. When this filter is specified directory operations are not monitored.
first_read	boolean	Filter the client requests for the first-read.
first_write	boolean	Filter the client requests for the first-write.
monitor_ads	boolean	Filter the client request for alternate data stream.
offline_bit	boolean	Filter the client request for offline bit set. FPolicy server receives notification only when offline files are accessed.
open_with_delete_intent	boolean	Filter the client request for open with delete intent.
open_with_write_intent	boolean	Filter the client request for open with write intent.
setattr_with_access_time_change	boolean	Filter the client setattr requests for changing the access time of a file or directory.
setattr_with_allocation_size_change	boolean	Filter the client setattr requests for changing the allocation size of a file.
setattr_with_creation_time_change	boolean	Filter the client setattr requests for changing the creation time of a file or directory.

Name	Type	Description
setattr_with_dacl_change	boolean	Filter the client setattr requests for changing dacl on a file or directory.
setattr_with_group_change	boolean	Filter the client setattr requests for changing group of a file or directory.
setattr_with_mode_change	boolean	Filter the client setattr requests for changing the mode bits on a file or directory.
setattr_with_modify_time_change	boolean	Filter the client setattr requests for changing the modification time of a file or directory.
setattr_with_owner_change	boolean	Filter the client setattr requests for changing owner of a file or directory.
setattr_with_sacl_change	boolean	Filter the client setattr requests for changing sacl on a file or directory.
setattr_with_size_change	boolean	Filter the client setattr requests for changing the size of a file.
write_with_size_change	boolean	Filter the client request for write with size change.

svm

Name	Type	Description
uuid	string	SVM UUID

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update a specific FPolicy event configuration for an SVM

PATCH `/protocols/fpolicy/{svm.uuid}/events/{name}`

Introduced In: 9.6

Updates a specific FPolicy event configuration for an SVM. A cluster-level FPolicy event configuration cannot be modified for a data SVM through REST. When the file operations and filters fields are modified, the previous values are retained and new values are added to the list of previous values. To remove a particular file operation or filter, set its value to false in the request.

Related ONTAP commands

- `fpolicy policy event modify`

Learn more

- DOC /protocols/fpolicy/{svm.uuid}/events

Parameters

Name	Type	In	Required	Description
name	string	path	True	
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Request Body

Name	Type	Description
file_operations	file_operations	Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.
filters	filters	Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.
monitor_fileop_failure	boolean	Specifies whether failed file operations monitoring is required.
name	string	Specifies the name of the FPolicy event.
protocol	string	Protocol for which event is created. If you specify protocol, then you must also specify a valid value for the file operation parameters. The value of this parameter must be one of the following: • cifs - for the CIFS protocol. • nfsv3 - for the NFSv3 protocol. • nfsv4 - for the NFSv4 protocol.
svm	svm	
volume_monitoring	boolean	Specifies whether volume operation monitoring is required.

Example request

```
{
  "name": "event_cifs",
  "protocol": "cifs",
  "svm": {
    "uuid": "string"
  }
}
```

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

Error Code	Description
9764873	The event is a cluster event
9764929	The file operation is not supported by the protocol
9764955	The filter is not supported by the protocol
9764930	The filter is not supported by any of the file operations
9764946	The protocol is specified without file operation or a file operation and filter pair
9765048	The monitor fileop failure option is set without protocol and file operations

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

file_operations

Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.

Name	Type	Description
access	boolean	Access operations
close	boolean	File close operations
create	boolean	File create operations
create_dir	boolean	Directory create operations
delete	boolean	File delete operations
delete_dir	boolean	Directory delete operations
getattr	boolean	Get attribute operations
link	boolean	Link operations
lookup	boolean	Lookup operations
open	boolean	File open operations
read	boolean	File read operations
rename	boolean	File rename operations
rename_dir	boolean	Directory rename operations
setattr	boolean	Set attribute operations
symlink	boolean	Symbolic link operations
write	boolean	File write operations

filters

Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.

Name	Type	Description
close_with_modification	boolean	Filter the client request for close with modification.
close_with_read	boolean	Filter the client request for close with read.
close_without_modification	boolean	Filter the client request for close without modification.
exclude_directory	boolean	Filter the client requests for directory operations. When this filter is specified directory operations are not monitored.
first_read	boolean	Filter the client requests for the first-read.
first_write	boolean	Filter the client requests for the first-write.
monitor_ads	boolean	Filter the client request for alternate data stream.
offline_bit	boolean	Filter the client request for offline bit set. FPolicy server receives notification only when offline files are accessed.
open_with_delete_intent	boolean	Filter the client request for open with delete intent.
open_with_write_intent	boolean	Filter the client request for open with write intent.
setattr_with_access_time_change	boolean	Filter the client setattr requests for changing the access time of a file or directory.
setattr_with_allocation_size_change	boolean	Filter the client setattr requests for changing the allocation size of a file.
setattr_with_creation_time_change	boolean	Filter the client setattr requests for changing the creation time of a file or directory.

Name	Type	Description
setattr_with_dacl_change	boolean	Filter the client setattr requests for changing dacl on a file or directory.
setattr_with_group_change	boolean	Filter the client setattr requests for changing group of a file or directory.
setattr_with_mode_change	boolean	Filter the client setattr requests for changing the mode bits on a file or directory.
setattr_with_modify_time_change	boolean	Filter the client setattr requests for changing the modification time of a file or directory.
setattr_with_owner_change	boolean	Filter the client setattr requests for changing owner of a file or directory.
setattr_with_sacl_change	boolean	Filter the client setattr requests for changing sacl on a file or directory.
setattr_with_size_change	boolean	Filter the client setattr requests for changing the size of a file.
write_with_size_change	boolean	Filter the client request for write with size change.

svm

Name	Type	Description
uuid	string	SVM UUID

fpolicy_event

The information that a FPolicy process needs to determine what file access operations to monitor and for which of the monitored events notifications should be sent to the external FPolicy server.

Name	Type	Description
file_operations	file_operations	Specifies the file operations for the FPolicy event. You must specify a valid protocol in the protocol parameter. The event will check the operations specified from all client requests using the protocol.
filters	filters	Specifies the list of filters for a given file operation for the specified protocol. When you specify the filters, you must specify the valid protocols and a valid file operations.
monitor_fileop_failure	boolean	Specifies whether failed file operations monitoring is required.
name	string	Specifies the name of the FPolicy event.
protocol	string	Protocol for which event is created. If you specify protocol, then you must also specify a valid value for the file operation parameters. The value of this parameter must be one of the following: • cifs - for the CIFS protocol. • nfsv3 - for the NFSv3 protocol. • nfsv4 - for the NFSv4 protocol.
svm	svm	
volume_monitoring	boolean	Specifies whether volume operation monitoring is required.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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